Simulation Arena

Deal Approval & Margin Balancing

Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $47,335.60M of expected profit.

Seed 42 | Run run_450ca2e3ab98 | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 22 Sep 2026, 08:23

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
0%
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
$6,029.07M
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
0%
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $468.44M, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
2%
At 2% the expected profit falls $2,355.77M below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $2,355.77M

EXPECTED PROFIT IF YOU CAVE IN
-$41,306.53M
What 25% actually earns you.
vs -$47,335.60M vs the optimum

PROFIT GIVEN AWAY
$47,335.60M
Buying +49 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
36.9%
The chance the deal lands.

WORST CASE (1-IN-20)
$0
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $2,580.24M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $241,763.99M on $258,023.80M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 17% discount is where they are most persuadable.

Evidence: Curve source: 250000 of your past deals

3. The sweet spot

0% discount gives the highest expected profit at $6,029.07M, with a 36.9% chance of winning.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

0% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $2,355.77M.

Evidence: margin of error $468.44M; safe range 0%

5. What the requested discount really costs

Moving from 0% to 25% buys +49 points of win probability but costs $47,335.60M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: -$41,306.53M vs $6,029.07M

Charts

Where profit peaks

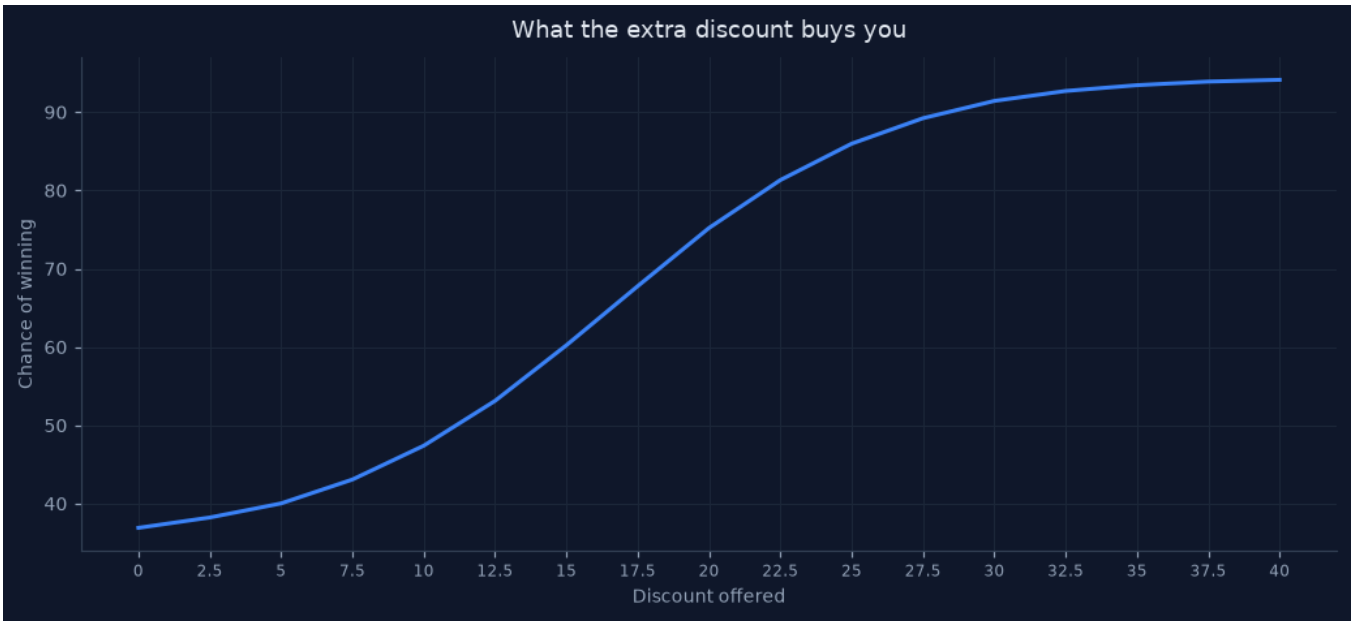
Everything from 0% gives the same answer within the model's margin of error; the peak is 0%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 0% range. Past that you are genuinely paying for the win.

What the extra discount buys you

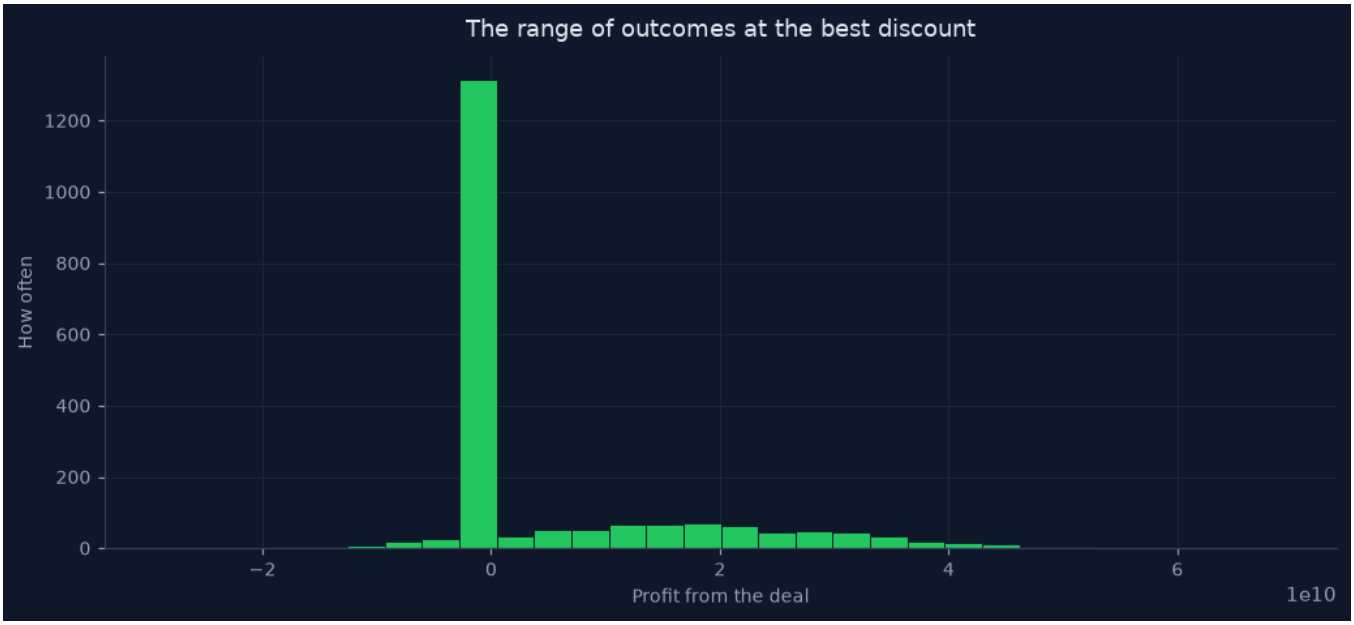
Curve learned from 250000 of your past deals



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 0% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$258,023,799,539	$16,233,267,836	36.9%	$6,029,073,471	$0	85% chance
2.5%	$251,573,204,550	$9,944,239,913	38.2%	$3,673,303,513	-$6,401,882,762	92% chance
5.0%	$245,122,609,562	$3,347,180,896	40.0%	$1,317,796,994	-$13,606,229,337	96% chance
7.5%	$238,672,014,573	-$3,181,880,436	43.1%	-$1,337,185,796	-$20,571,497,041	99% chance
10.0%	$232,221,419,585	-$9,603,830,075	47.4%	-$4,439,556,827	-$26,716,214,076	100% chance
12.5%	$225,770,824,596	-$15,564,103,673	53.1%	-$8,215,753,883	-$34,137,409,959	100% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$219,320,229,608	-$22,364,842,382	60.2%	-$13,553,397,154	-$42,622,707,211	100% chance
17.5%	$212,869,634,619	-$28,535,235,985	67.8%	-$19,285,168,618	-$49,534,041,418	100% chance
20.0%	$206,419,039,631	-$35,172,663,991	75.2%	-$26,465,912,894	-$57,219,980,776	100% chance
22.5%	$199,968,444,642	-$41,738,672,004	81.3%	-$34,080,455,399	-$64,204,132,025	100% chance
25.0%	$193,517,849,654	-$48,271,378,903	86.0%	-$41,306,530,463	-$71,013,645,738	100% chance
27.5%	$187,067,254,666	-$54,817,717,816	89.2%	-$48,768,672,238	-$77,687,201,468	100% chance
30.0%	$180,616,659,677	-$61,488,156,064	91.4%	-$56,249,920,531	-$85,120,996,822	100% chance
32.5%	$174,166,064,689	-$67,673,036,766	92.7%	-$62,732,196,910	-$91,402,475,680	100% chance
35.0%	$167,715,469,700	-$74,283,109,327	93.4%	-$69,531,897,234	-$97,514,608,541	100% chance
37.5%	$161,264,874,712	-$80,270,645,615	93.9%	-$75,355,523,576	-$103,348,680,397	100% chance
40.0%	$154,814,279,723	-$86,958,410,319	94.1%	-$81,900,569,472	-$110,279,284,830	100% chance

Row 1 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (0%)	Asked (25%)	Difference
Discount	0.0%	25.0%	+25.0 pts
Win chance	36.9%	86.0%	+49.1 pts
Profit if won	$16,233,267,836	-$48,271,378,903	-$64,504,646,740
Expected profit	$6,029,073,471	-$41,306,530,463	-$47,335,603,933
Range where the answer is the same	0%	outside it	margin of error $468.44M

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	$6,029,073,471	$0	Inside	safe to approve
2.5%	$3,673,303,513	$2,355,769,957	Outside	costs you $2,355.77M
5.0%	$1,317,796,994	$4,711,276,476	Outside	costs you $4,711.28M
7.5%	-$1,337,185,796	$7,366,259,266	Outside	costs you $7,366.26M
10.0%	-$4,439,556,827	$10,468,630,297	Outside	costs you $10,468.63M
12.5%	-$8,215,753,883	$14,244,827,354	Outside	costs you $14,244.83M
15.0%	-$13,553,397,154	$19,582,470,624	Outside	costs you $19,582.47M
17.5%	-$19,285,168,618	$25,314,242,089	Outside	costs you $25,314.24M
20.0%	-$26,465,912,894	$32,494,986,364	Outside	costs you $32,494.99M
22.5%	-$34,080,455,399	$40,109,528,870	Outside	costs you $40,109.53M
25.0%	-$41,306,530,463	$47,335,603,933	Outside	costs you $47,335.60M
27.5%	-$48,768,672,238	$54,797,745,709	Outside	costs you $54,797.75M
30.0%	-$56,249,920,531	$62,278,994,001	Outside	costs you $62,278.99M
32.5%	-$62,732,196,910	$68,761,270,381	Outside	costs you $68,761.27M
35.0%	-$69,531,897,234	$75,560,970,704	Outside	costs you $75,560.97M
37.5%	-$75,355,523,576	$81,384,597,047	Outside	costs you $81,384.60M
40.0%	-$81,900,569,472	$87,929,642,942	Outside	costs you $87,929.64M

Highlighted rows are the range where the answer is the same (0%) - within about $468.44M of the best.

Recommended discount per deal (top 25)

The same logic applied to each deal individually.

Deal	Customer	Value	Margin	Best discount	Expected profit
Q-000116895	Acct-010340	$24,000,000	13.6%	0%	$1,200,831
Q-000175180	Acct-210762	$24,000,000	-4.2%	0%	-$370,441
Q-000293851	Acct-181197	$24,000,000	6.4%	0%	$564,523
Q-000153461	Acct-122055	$24,000,000	30.6%	0%	$2,709,181
Q-000420606	Acct-175726	$24,000,000	18.4%	0%	$1,628,876
Q-000161367	Acct-017235	$24,000,000	17.4%	0%	$1,537,595
Q-000227845	Acct-192964	$24,000,000	10.8%	0%	$960,665
Q-000400243	Acct-180720	$24,000,000	19.1%	0%	$1,694,456
Q-000070031	Acct-176770	$24,000,000	14.4%	0%	$1,277,932
Q-000223484	Acct-189715	$24,000,000	5.3%	0%	$471,470
Q-000056573	Acct-025370	$24,000,000	0.8%	0%	$69,125
Q-000224794	Acct-110864	$24,000,000	21.8%	0%	$1,927,533
Q-000451216	Acct-078030	$24,000,000	-0.2%	0%	-$16,838
Q-000358069	Acct-051316	$24,000,000	-3.2%	0%	-$283,591
Q-000106713	Acct-095976	$24,000,000	17.3%	0%	$1,534,050
Q-000291780	Acct-144354	$24,000,000	4.2%	0%	$369,555
Q-000017149	Acct-168647	$24,000,000	16.5%	0%	$1,458,721
Q-000266451	Acct-223251	$24,000,000	-8.0%	0%	-$710,750
Q-000467907	Acct-116435	$24,000,000	0.4%	0%	$37,221
Q-000284838	Acct-072965	$24,000,000	20.2%	0%	$1,789,282
Q-000361102	Acct-064034	$24,000,000	33.0%	0%	$2,923,646
Q-000057013	Acct-121440	$24,000,000	32.3%	0%	$2,863,383
Q-000135049	Acct-047493	$24,000,000	29.0%	0%	$2,569,158
Q-000090928	Acct-225367	$24,000,000	3.4%	0%	$300,429
Q-000249451	Acct-084588	$24,000,000	6.1%	0%	$542,368

What we recommend

Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $47,335.60M of expected profit.

Risk High 100 out of 100 | Confidence: High

Why

• Only 0% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 0% ($6,029.07M).

• The requested 25% delivers -$41,306.53M - $47,335.60M less than the peak, which is more than the model's margin of error.

• Win probability only improves +49 points for that giveaway.

What to do

1. Approve 0% as the standard concession - it is the only discount the model can tell apart from the rest.

2. Above 0% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $1,893.42M per point of discount.

3. At 25% there is a 100% chance of breaching the 12.0% margin floor - add a volume or term commitment to earn the deeper discount.

The trade-off: You are trading $47,335.60M of expected profit for +49 points of win probability.

What we assumed and what to watch

Worth knowing before you act

• At 25% discount there is a 100% chance the deal actually costs you money to deliver even if you win it.

• Your file holds 500,000 rows. We are running on an evenly spread sample of 250,000 of them (50.0%), keeping rows intact rather than splitting a record down the middle, because a working set that big does not fit in memory while keeping the tool responsive. Every number below comes from those 250,000 rows.

Assumptions

• List value $258,023,799,539; cost to deliver $233,640,482,791 at the 9.5% standard margin your file implies, plus $8,123,509,942 of consumption effects on top, giving a total cost base of $241,763,992,733 and an effective 6.3% margin once those effects are included.

• Consumption effects: 0.2% of value is consumed as unmetered burst you still pay for; 4 months of rollover defers revenue.

• Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 16.6% discount (source: 250000 of your past deals).

• Win probability uncertainty: plus or minus 5 points.

• Cost volatility: plus or minus 6% of cost.

• Margin floor: 12.0%.

• Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $468.44M at the best one), or smaller than 1% of the best profit - whichever is larger.

• The range where the answer is the same is 0%, taken from the 17 discount levels we tested.

• 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).

• Deal / quote ID -> your 'deal_id' column

• Customer name -> your 'customer' column

• Deal value (list price) -> your 'deal_value' column

• Cost to deliver -> your 'cost' column

• Standard margin (%) -> your 'margin_pct' column

• Discount given (%) -> your 'discount_pct' column

• Won? (1/0) -> your 'won' column

• Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column

• Unmetered burst used -> your 'burst_allowance_pct' column

• Rollover window -> your 'rollover_window_months' column

• Data used: 250,000 of the 500,000 rows in your file (50.0%), drawn evenly across it.

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
1.42s	prepare	Won? (1/0) -> your 'won' column
1.42s	prepare	Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
1.42s	prepare	Unmetered burst used -> your 'burst_allowance_pct' column

Time	Stage	What happened
1.42s	prepare	Rollover window -> your 'rollover_window_months' column
9.19s	prepare	Read 250,000 records into the model.
9.19s	simulate	Running the simulation…
9.22s	read	Reading the deal you gave us...
9.34s	read	Found 250000 deal(s) worth $258,023.80M at an average 9.5% margin.
9.34s	prepare	Cost to deliver is about $233,640.48M.
9.54s	prepare	Added the consumption effects from your file: $8,123.51M of extra cost to serve, taking the cost base to $241,763.99M.
9.54s	prepare	Working out how much a discount really moves the buyer...
23.76s	prepare	Learned the buyer's behaviour from 250000 past deals (they flip hardest around 16.6% discount).
23.76s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...
23.76s	simulate	Simulated 11% of the discount levels...
23.76s	simulate	Simulated 23% of the discount levels...
23.77s	simulate	Simulated 35% of the discount levels...
23.77s	simulate	Simulated 47% of the discount levels...
23.77s	simulate	Simulated 58% of the discount levels...
23.77s	simulate	Simulated 70% of the discount levels...
23.77s	simulate	Simulated 82% of the discount levels...
23.77s	simulate	Simulated 94% of the discount levels...
23.78s	simulate	Simulated 100% of the discount levels...
23.78s	summarize	All discount levels tested. Looking for the sweet spot...
23.78s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 0%.
23.94s	summarize	Done. Best discount is 0% ($6,029.07M expected profit); the range where the answer is the same is 0%.